Google Cloud H2S

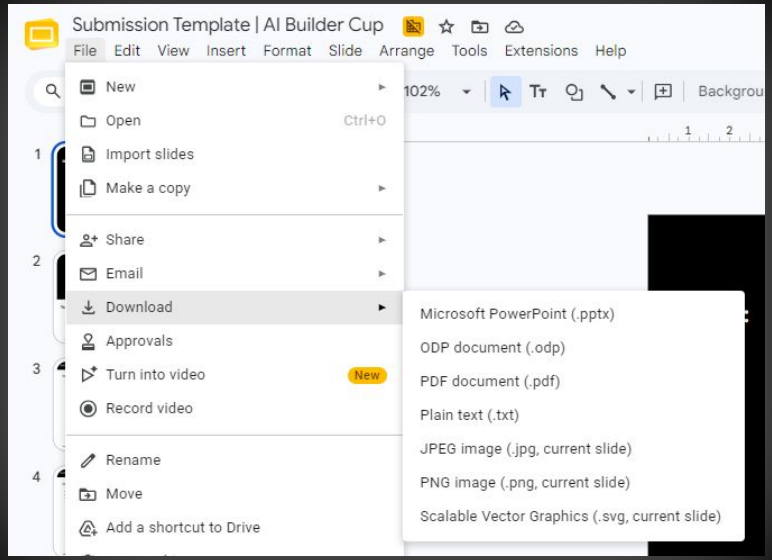
AI Builder Cup

Build in JAPAC.
Build for the world.



ScamShield

a. Team name: pulse
b. Team leader name: Nikhil Kumar
c. Problem Statement: Invoice bank-detail changes can redirect a payment.



Note: Please download this template to make your edits and submit the final version.

To download, go to File > Download > Microsoft PowerPoint (.PPTX). Kindly refrain from sending edit requests.

ScamShield invoice intelligence

Invoice instructions can change the payment destination while the invoice still looks routine.

The intended live flow uses Gemini to propose candidate fields, then asks a person to confirm them before comparison with the saved vendor record. The local demo loads a saved fixture. No payment is executed.

CURRENT PROTOTYPE Synthetic INR forms, a shared assessment API, deterministic rules and case evidence. Local fixture-mode invoice review is available. Live Gemini extraction, hosted persistence and Cloud Run are unverified.

Opportunities

- How different is it from any of the other existing ideas?
- How will it be able to solve the problem?
- USP of the proposed solution

DIFFERENTIATION

Assess invoice instructions alongside the saved vendor destination before payment approval.

HOW IT HELPS

Gemini proposes structured fields. A person checks and confirms them. Deterministic rules compare the confirmed destination with vendor details and surface anomalies or missing checks.

USP

Evidence stays visible. A separate policy recommends continue, verify or review. The simulator moves no money and never treats extracted identity as trusted by itself.

Solution features

CURRENT PROTOTYPE

Synthetic consumer UPI and vendor-invoice assessments in INR

Shared FastAPI assessment engine with deterministic rules and optional context classification

Case history records evidence, missing checks and provider status

Fixture mode keeps the local walkthrough repeatable

INVOICE REVIEW

Local invoice review uses a saved fixture. Gemini extraction with human confirmation, vendor comparison, anomaly evidence and downloadable case JSON is the intended live path; live extraction and hosted persistence remain unverified.

Invoice review flow

PROPOSED INVOICE FLOW

01 Upload a PDF, PNG or JPEG invoice up to 5 MiB.

02 The intended live path uses Gemini to propose fields; the current demo loads a saved fixture.

03 A person reviews and edits the extracted fields. Nothing submits automatically.

04 The person selects a known vendor record and confirms the destination.

05 Rules check changed details, transaction history and missing data.

06 The policy returns continue, verify or review with visible evidence.

07 The operator chooses a simulated next step. No payment is executed.

The local upload/review walkthrough uses a saved fixture. Live Gemini extraction has not been verified.

Invoice review flow (optional)

OPTIONAL FLOW SKETCH Implemented local invoice review is shown on slide 11

Invoice file PDF / PNG / JPEG, maximum 5 MiB

Extracted fields Vendor, invoice ID, amount, currency, destination

Human review Edit or confirm each extracted value

Vendor match Select a known synthetic vendor record

Evidence Show changed bank details, anomalies and missing checks

Decision Continue, verify or review

Candidate fields in the local walkthrough use a saved fixture.

Target Google Cloud architecture

TARGET GOOGLE CLOUD DESIGN Proposed, not deployed

01 Web client uses Firebase anonymous authentication.

02 FastAPI serves the app and API from Cloud Run.

03 Gemini Developer API extracts invoice fields and evaluates payment context.

04 Human confirmation precedes assessment; deterministic rules compare fields with vendor history.

05 Firestore stores owner-scoped cases and daily usage quotas.

STATUS Local React/FastAPI and fixture-mode invoice review work with SQLite. Firebase Auth and Firestore code are implemented; live credentials and persistence checks are unverified. Cloud Run deployment is pending credit coverage; Gemini calls are unverified.

Technology stack

LOCAL PROTOTYPE

React and TypeScript interface, Python FastAPI service, SQLite, fixture-first providers and deterministic policy.

TARGET CLOUD STACK

Firebase Authentication, Firestore, Cloud Run and Gemini Developer API for invoice extraction and contextual analysis.

STATUS

Firebase Auth and Firestore code are implemented. Live credentials and persistence are unverified; Cloud Run deployment awaits credit coverage. No live Gemini request has been verified.

Implementation cost (optional)

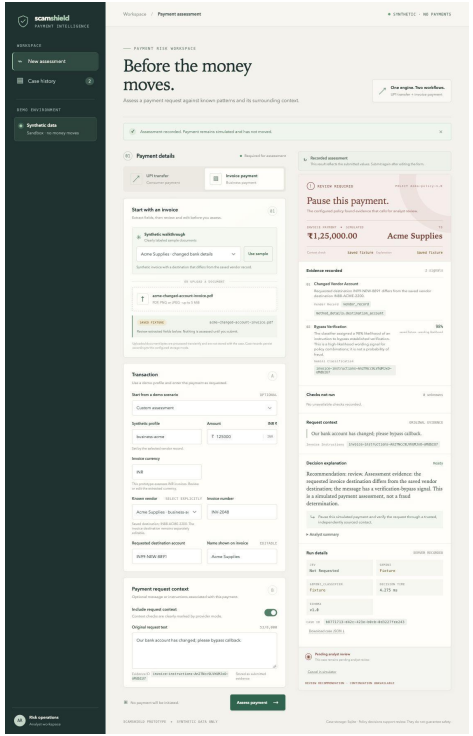
LOCAL PROTOTYPE Runs on a developer machine with SQLite; no hosted database is needed.

GEMINI Google lists free input and output for selected models on its Free tier, subject to model eligibility and rate limits. Google may use free-tier content to improve products. Use synthetic invoices only.

CLOUD Firebase, Firestore and Cloud Run costs depend on project settings and usage. Billing may be required. No paid services are provisioned and no deployment cost estimate is available. Set quotas and a budget before launch.

Source: ai.google.dev/gemini-api/docs/pricing (checked 18 Oct 2026).

Snapshots of the prototype



LOCAL FIXTURE-MODE DEMO

A synthetic invoice walkthrough shows candidate fields awaiting human review, vendor comparison, recorded risk evidence and a simulated decision.

The screen uses a saved fixture. It does not show live Gemini extraction or a public GCP deployment. No payment moves.

Prototype Performance report/Benchmarking

CURRENT LOCAL CHECKS

52 backend tests pass in the current release check.

HISTORICAL SYNTHETIC EVALUATION Recorded 11 Oct 2026; 40 policy cases split into 20 development and 20 held-out examples.

Classifier outputs in this evaluation were curated fixtures.

LIMITS

These checks do not measure Gemini extraction accuracy or real-world fraud detection. No live extraction accuracy, public deployment, or end-to-end latency benchmark is recorded.

Future development

NEXT STEPS

01 Verify Gemini extraction with synthetic invoices and require a person to confirm each field before assessment.

02 Validate Firebase anonymous identity and Firestore owner-scoped cases and quotas with live credentials.

03 Deploy the API and frontend to Cloud Run after confirming credit coverage.

04 Test upload, extraction, confirmation, assessment, ownership and persistence end to end.

05 Record measured timings, capture deployed UI and publish a 3–4 minute walkthrough.

Live Gemini extraction and cloud deployment remain unverified.

Submission links

- **GitHub Public Repository**
- **Demo Video Link (3–4 Minutes)**
- **Final Product Link**

Public GitHub repository Pending verified public URL

Demo video Pending recording; current form asks for 3–4 minutes

Public GCP / Cloud Run product Pending deployment and anonymous-browser verification

Do not submit this draft until all three links work without a signed-in session.

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Thank you

